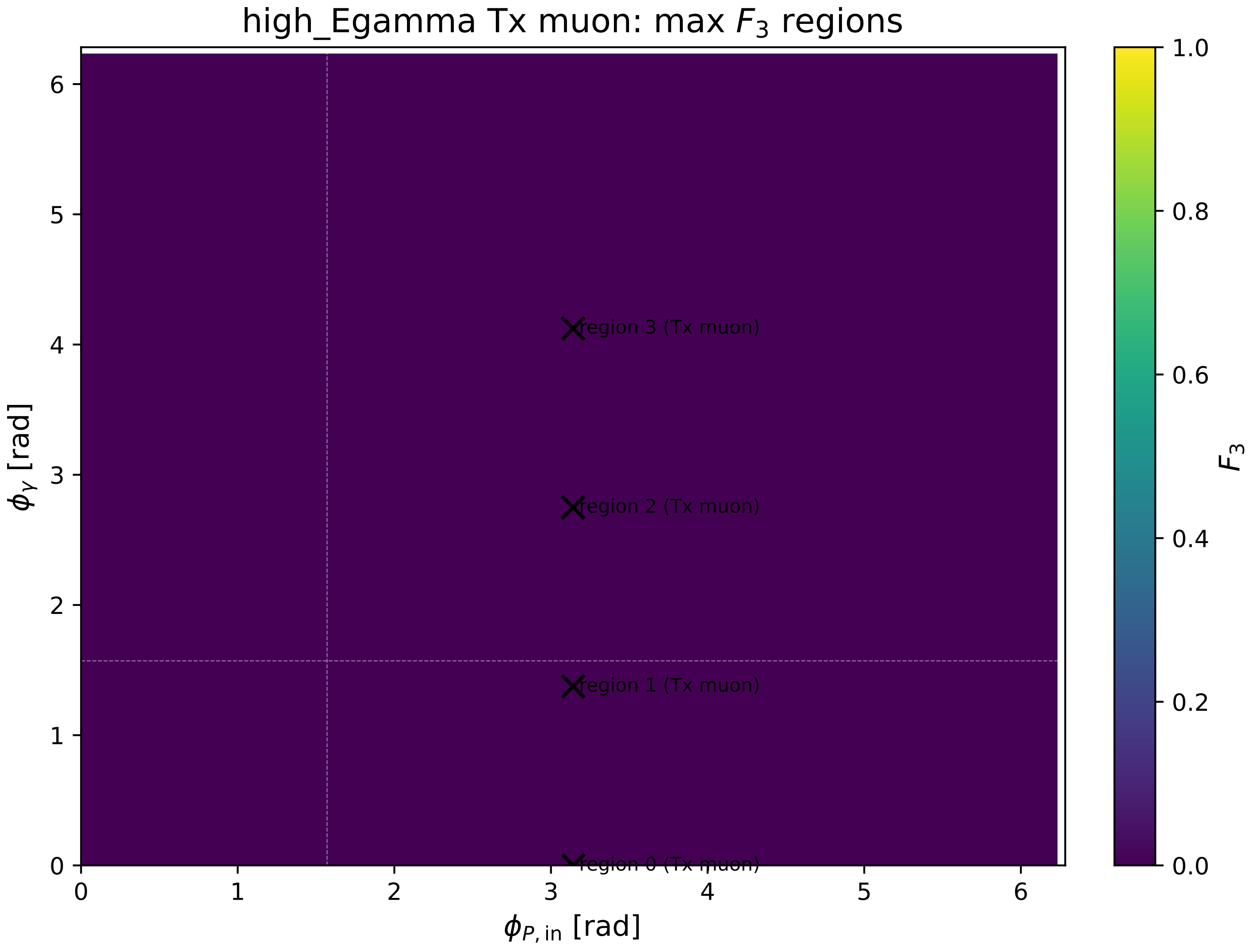
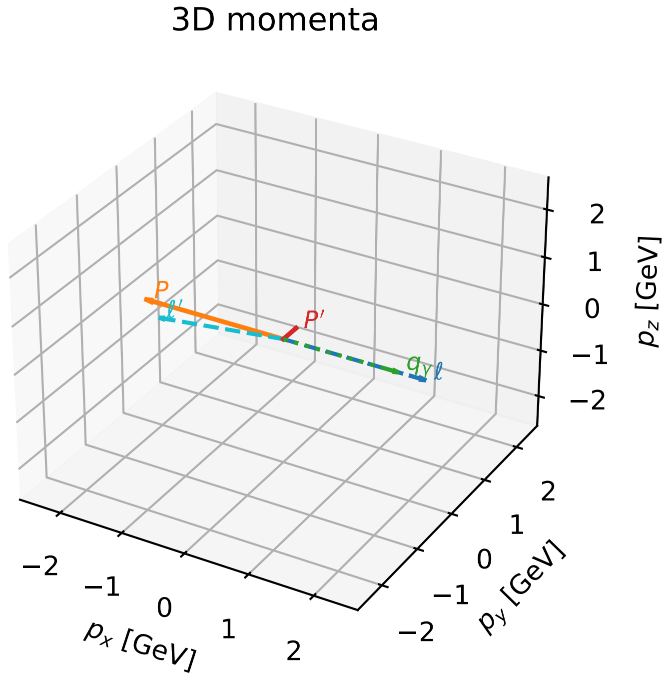


high_Egamma Tx muon: max F_3 regions
region 3 (Tx muon)
region 2 (Tx muon)
region 1 (Tx muon)
region 0 (Tx muon)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
F_3
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

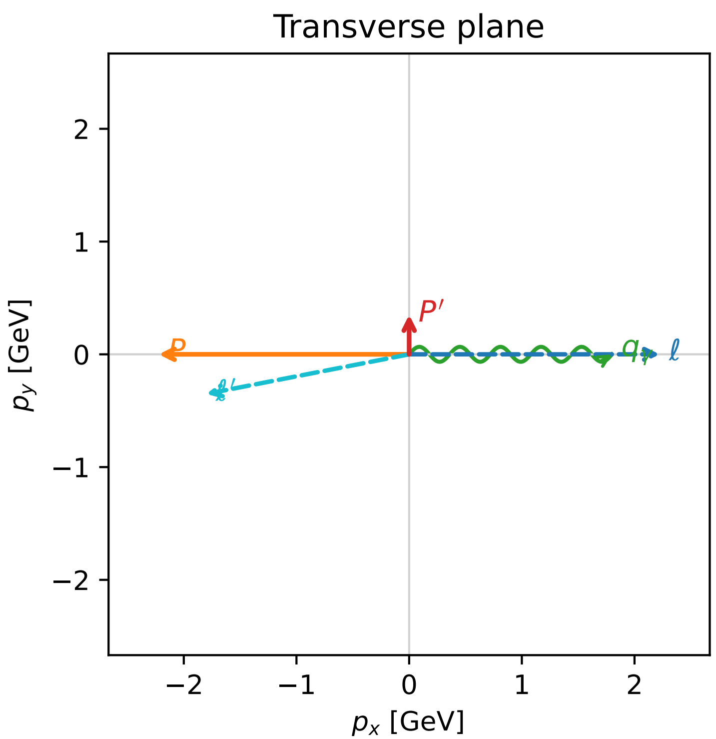
Tx muon characteristic max F_3 configuration

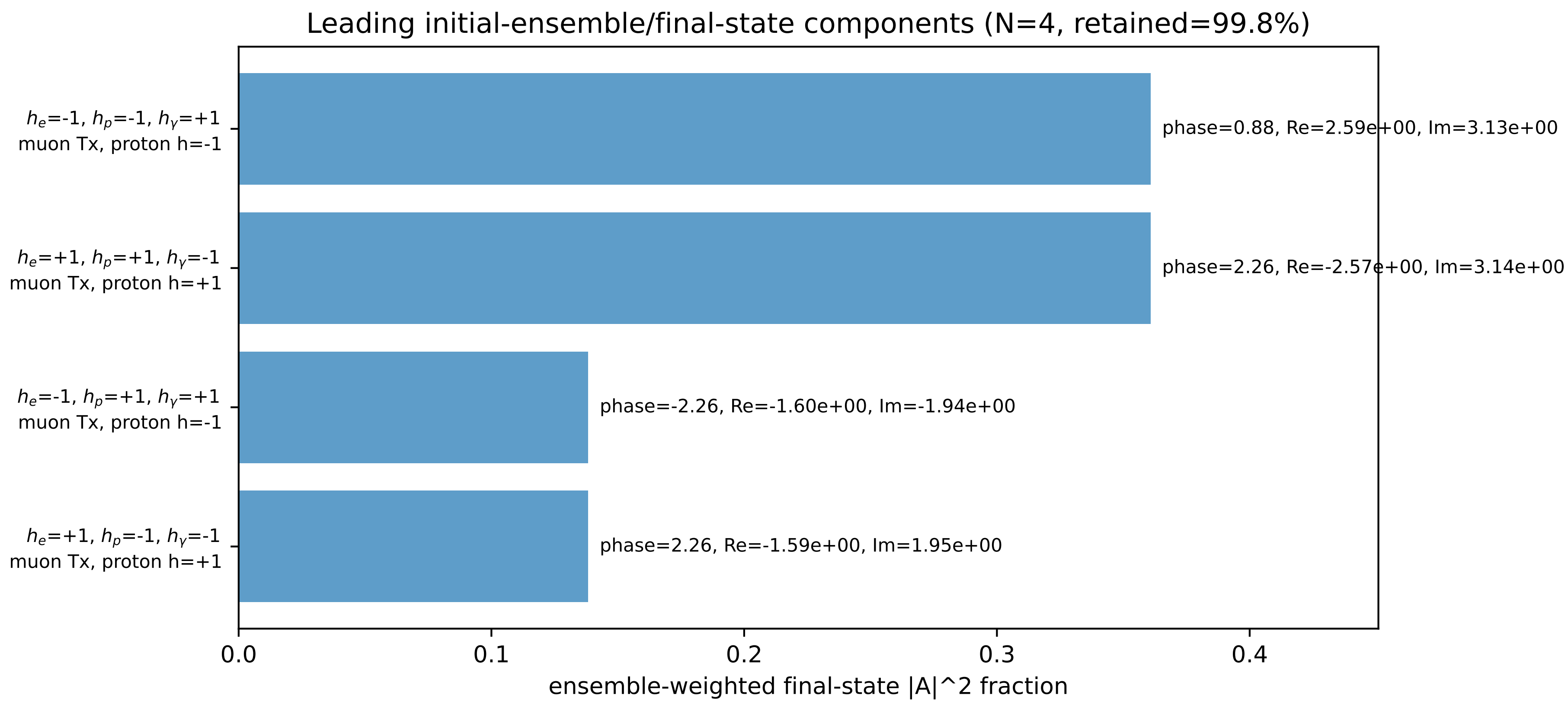


f3_high_egamma_tx_electron_region_0 F_3=0
region: high_Egamma_Tx_electron_region_0
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\mu}, h_p)$

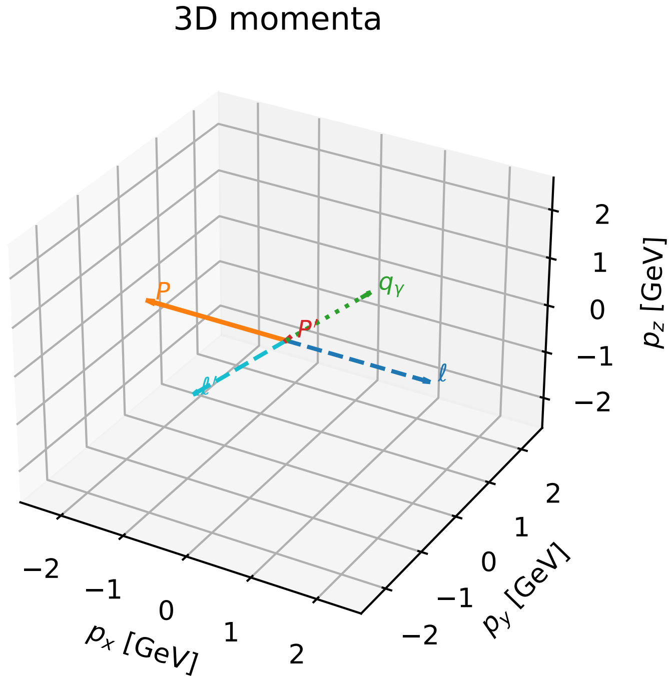
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.351095
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=0
Q^2=16.1576, x_B=0.81756, t=-3.07361, W^2=4.48544, y=0.958225
$\theta(\ell', \gamma)$=168.963 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 2.2231 p_y= -0.0000 p_z= -0.0000
P E= 2.4129 p_x= -2.2231 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8370 p_x= -1.8000 p_y= -0.3511 p_z= 0.0000
P' E= 1.0016 p_x= 0.0000 p_y= 0.3511 p_z= 0.0000
q_γ E= 1.8000 p_x= 1.8000 p_y= 0.0000 p_z= 0.0000





Tx muon characteristic max F_3 configuration

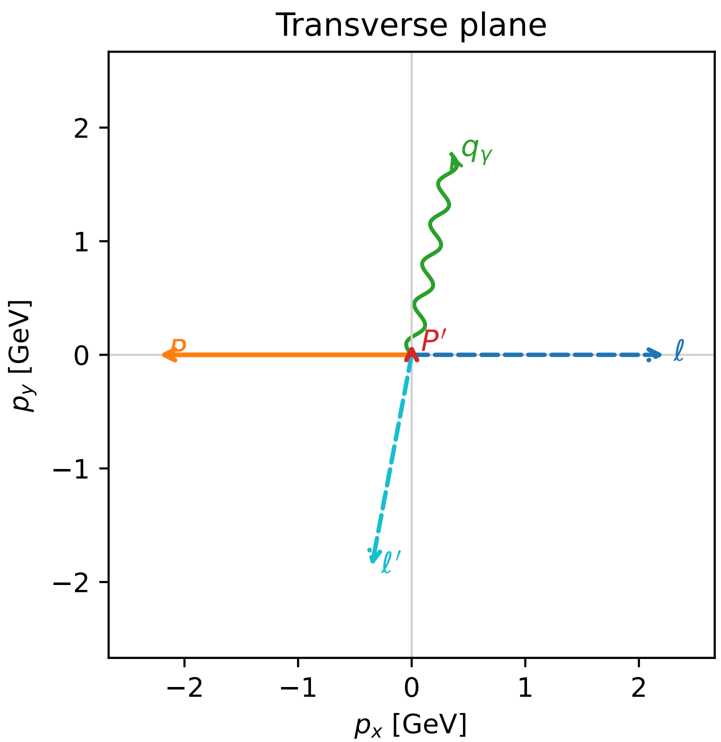


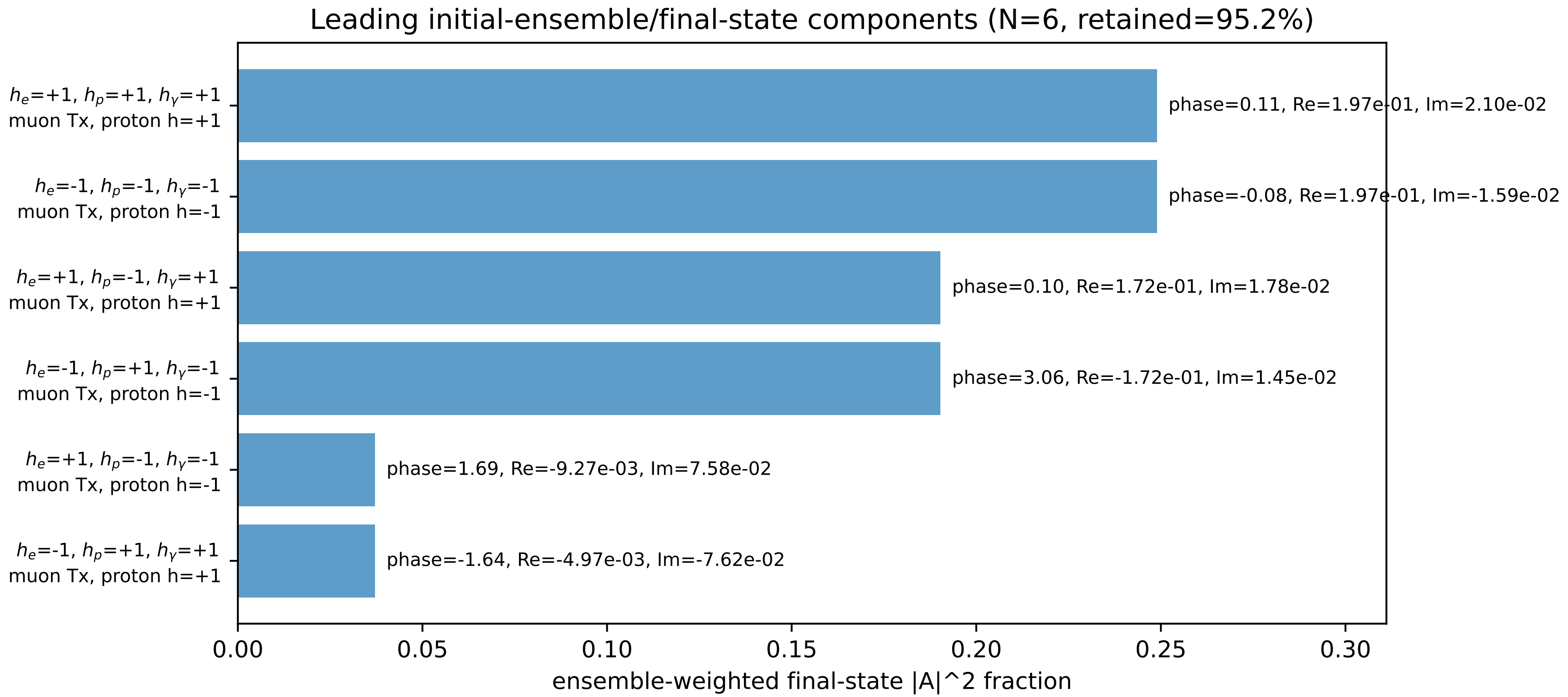

```
f3_high_egamma_tx_electron_region_1  F_3=0
region: high_Egamma_Tx_electron_region_1
outgoing-state purity: 0.500018
kinematic point: high_s_high_theta_in_high_Egamma
```

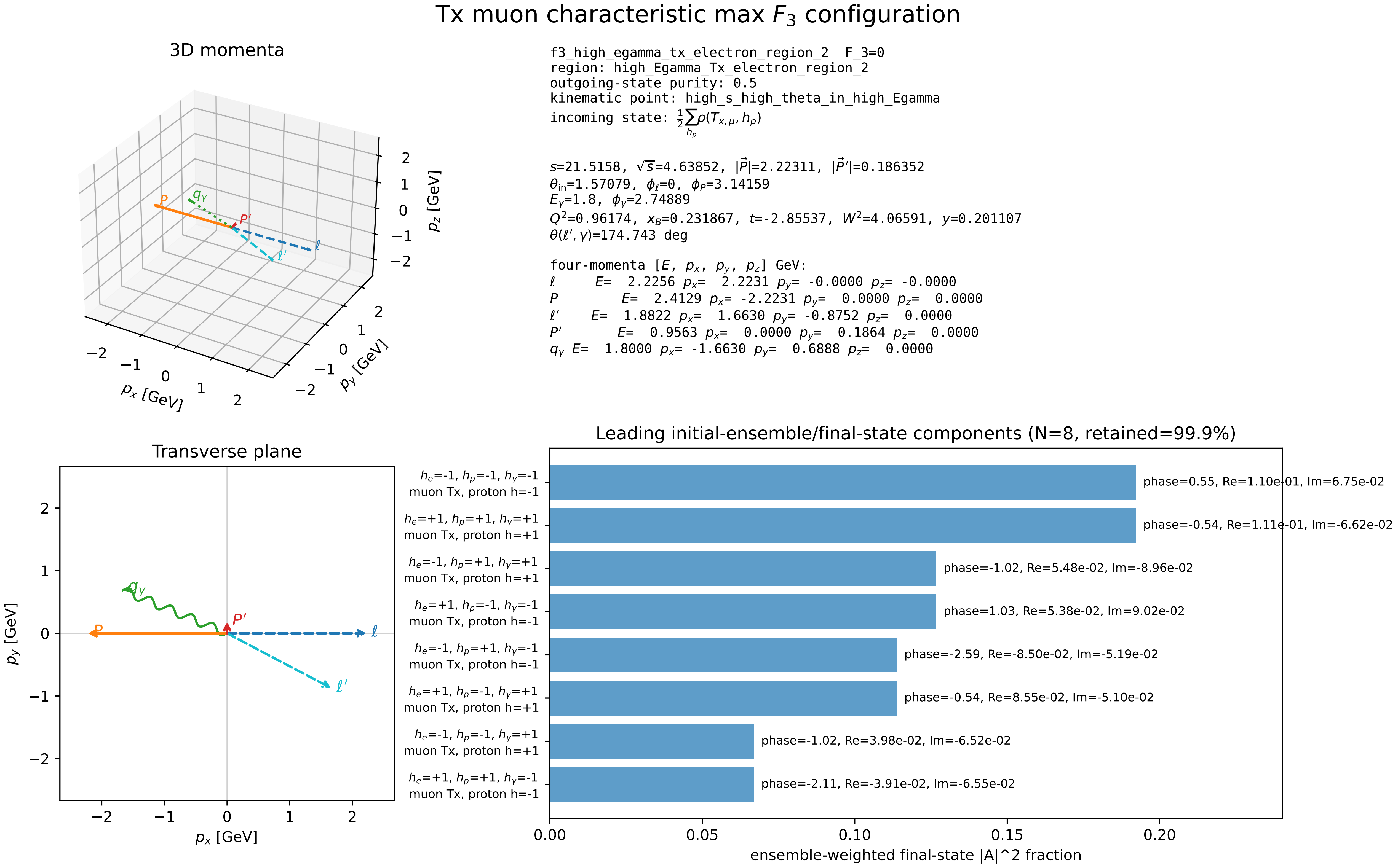
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\mu}, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.0945448
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=1.37445
Q^2=9.97753, x_B=0.765288, t=-2.78984, W^2=3.93993, y=0.632132
$\theta(\ell', \gamma)$=179.442 deg

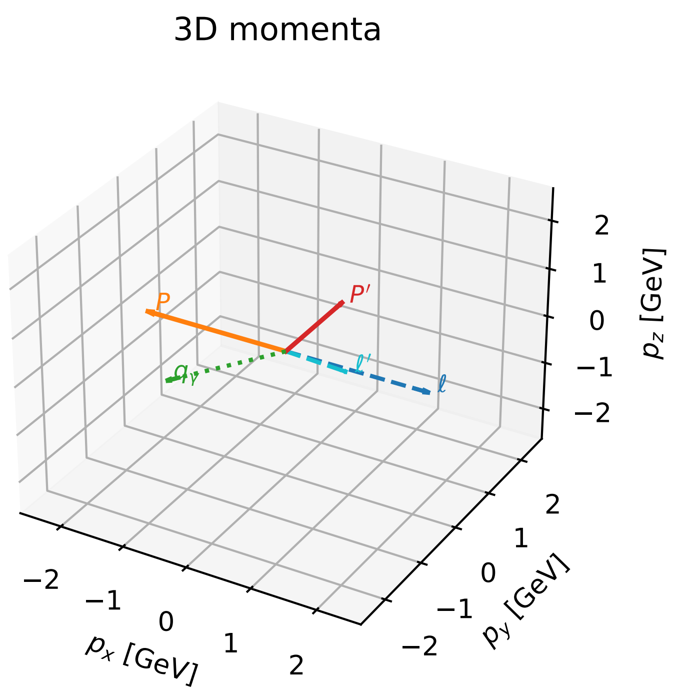
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 2.2231 p_y= -0.0000 p_z= -0.0000
P E= 2.4129 p_x= -2.2231 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8958 p_x= -0.3512 p_y= -1.8600 p_z= 0.0000
P' E= 0.9428 p_x= 0.0000 p_y= 0.0945 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.3512 p_y= 1.7654 p_z= 0.0000






Tx muon characteristic max F_3 configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
q_γ
P'
ℓ
ℓ'
f3_high_egamma_tx_electron_region_2 F_3=0
region: high_Egamma_Tx_electron_region_2
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\mu}, h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=0.186352
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=2.74889
Q^2=0.96174, x_B=0.231867, t=-2.85537, W^2=4.06591, y=0.201107
$\theta(\ell', \gamma)$=174.743 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 2.2231 p_y= -0.0000 p_z= -0.0000
P E= 2.4129 p_x= -2.2231 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8822 p_x= 1.6630 p_y= -0.8752 p_z= 0.0000
P' E= 0.9563 p_x= 0.0000 p_y= 0.1864 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.6630 p_y= 0.6888 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=8, retained=99.9%)
h_e=-1, h_p=-1, h_γ=-1 muon Tx, proton h=-1
phase=0.55, Re=1.10e-01, Im=6.75e-02
h_e=+1, h_p=+1, h_γ=+1 muon Tx, proton h=+1
phase=-0.54, Re=1.11e-01, Im=-6.62e-02
h_e=-1, h_p=+1, h_γ=+1 muon Tx, proton h=+1
phase=-1.02, Re=5.48e-02, Im=-8.96e-02
h_e=+1, h_p=-1, h_γ=-1 muon Tx, proton h=-1
phase=1.03, Re=5.38e-02, Im=9.02e-02
h_e=-1, h_p=+1, h_γ=-1 muon Tx, proton h=-1
phase=-2.59, Re=-8.50e-02, Im=-5.19e-02
h_e=+1, h_p=-1, h_γ=+1 muon Tx, proton h=+1
phase=-0.54, Re=8.55e-02, Im=-5.10e-02
h_e=-1, h_p=-1, h_γ=+1 muon Tx, proton h=+1
phase=-1.02, Re=3.98e-02, Im=-6.52e-02
h_e=+1, h_p=+1, h_γ=-1 muon Tx, proton h=-1
phase=-2.11, Re=-3.91e-02, Im=-6.55e-02
ensemble-weighted final-state |A|^2 fraction

Tx muon characteristic max F_3 configuration




```
f3_high_egamma_tx_electron_region_3  F_3=0
region: high_Egamma_Tx_electron_region_3
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{x,\mu}, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.57149
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=4.12334
Q^2=0.019849, x_B=0.00175464, t=-7.0722, W^2=12.1723, y=0.548479
$\theta(\ell', \gamma)$=119.47 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 2.2231 p_y= -0.0000 p_z= -0.0000
P E= 2.4129 p_x= -2.2231 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.0084 p_x= 1.0000 p_y= -0.0748 p_z= 0.0000
P' E= 1.8301 p_x= 0.0000 p_y= 1.5715 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.0000 p_y= -1.4966 p_z= 0.0000